Core Java Study Plan Day 1

Interview Questions

1. What is Java, and why is it considered platform independent?

2. Explain the difference between JVM, JRE, and JDK.

3. What is the role of the JVM in Java program execution?

4. What is Java bytecode, and why is it important?

5. Explain the complete flow from a Java source file to program execution.

6. What is the difference between JDK and JRE?

7. What are the major components of the JDK?

8. What are the main responsibilities of the JVM?

9. What is the Java compiler, and what does javac do?

10. What is the difference between source code, bytecode, and machine code?

11. Why can the same Java bytecode run on different operating systems?

12. What are the major features of Java?

13. What is the role of the class loader in the JVM?

14. What is the purpose of JVM memory areas?

15. What is JIT compilation, and how does it improve Java performance?

16. What is garbage collection in Java?

17. Explain the basic structure of a Java program.

18. What is the purpose of the main() method?

19. What is the difference between a class, object, method, and variable in a basic Java program?

20. Why is public static void main(String[] args) used as the entry point of a Java application?

MCQ Questions

1. Which component executes Java bytecode?

- A. JDK
- B. JVM
- C. JRE
- D. javac

2. Which command compiles a Java source file?

- A. java
- B. javac
- C. jvm
- D. jar

3. Which file extension is normally produced after compiling a Java source file?

- A. .java
- B. .exe
- C. .class
- D. .jar

4. Which component contains the JVM and core runtime libraries?

- A. JDK
- B. JRE
- C. javac
- D. JAR

5. Which component is primarily used by developers to develop Java applications?

- A. JDK
- B. JRE
- C. JVM
- D. JAR

6. What does JVM stand for?

- A. Java Variable Machine
- B. Java Virtual Machine
- C. Java Verified Machine
- D. Java Visual Machine

7. What does JDK stand for?

- A. Java Development Kit
- B. Java Deployment Kit
- C. Java Developer Kernel
- D. Java Data Kit

8. What does JRE stand for?

- A. Java Runtime Environment
- B. Java Resource Environment
- C. Java Runtime Engine
- D. Java Resource Engine

9. Which feature allows Java bytecode to run on multiple operating systems?

- A. Encapsulation
- B. Platform independence

- C. Inheritance
- D. Overloading

10. Which tool is used to start a compiled Java application?

- A. javac
- B. java
- C. javadoc
- D. jar

11. Which method is the traditional entry point of a Java application?

- A. start()
- B. run()
- C. main()
- D. execute()

12. Which keyword is used to declare a Java class?

- A. object
- B. class
- C. type
- D. define

13. Which of the following is a valid main method declaration?

- A. public void main()
- B. static public main(String args)
- C. public static void main(String[] args)
- D. public main(String[] args)

14. Which JVM component loads Java classes into memory?

- A. Garbage Collector
- B. Class Loader
- C. JIT Compiler
- D. Interpreter

15. What is JIT mainly associated with?

- A. Database management
- B. Runtime compilation
- C. Source code formatting
- D. Package creation

16. Which of the following is NOT a Java source file extension?

- A. .java
- B. .class
- C. .jav
- D. All are source extensions

17. Which statement best describes bytecode?

- A. Native machine code for one CPU
- B. Intermediate code executed by the JVM
- C. Java source code
- D. Operating system code

18. Which Java feature helps manage unused objects automatically?

- A. Inheritance
- B. Garbage collection
- C. Method overloading
- D. Encapsulation

19. Which of these is generally included in the JDK but not required merely to run Java programs?

- A. JVM
- B. Development tools
- C. Runtime libraries
- D. Class loader

20. Which sequence best represents basic Java execution?

- A. .class → .java → JVM
- B. .java → javac → .class → JVM
- C. JVM → .java → javac
- D. JDK → .java → operating system

Puzzles

1. The Three Boxes Puzzle

There are three boxes labeled:

APPLES

ORANGES

APPLES & ORANGES

All three labels are incorrect.

You are allowed to pick only one fruit from one box, without looking inside.

Question: From which box should you pick the fruit first to correctly identify all three boxes?

A) APPLES

B) ORANGES

C) APPLES & ORANGES

D) Cannot be determined

2. The Four People Puzzle

Four people — A, B, C and D — are standing in a queue.

A is not first.

B is immediately behind C.

D is somewhere ahead of A.

C is not first.

Question: Which of the following can be the correct order?

A) A C B D

B) C B D A

C) D C B A

D) B C D A

3. The Switches Puzzle

There are three switches outside a closed room. Inside the room is one bulb. Only one switch controls the bulb.

You may operate the switches as much as you want, but you can enter the room only once.

Question: How can you determine which switch controls the bulb?

A) Turn on one switch and immediately enter

B) Turn on one switch for some time, turn it off, turn on another switch, then enter and use the bulb's light/heat

C) Turn on all three switches and enter

D) It is impossible to determine

4. The Missing Number Puzzle

Find the next number:

2, 6, 12, 20, 30, 42, ?

A) 52

B) 54

C) 56

D) 60

5. The Two Doors Puzzle

You are standing before two doors.

One door leads to safety.

One door leads to danger.

One guard always tells the truth.

The other guard always lies.

You don't know which guard is which.

You can ask one question to one guard.

Question: What should you ask to identify the safe door?

A) Which door leads to safety?

B) Which door would the other guard say leads to safety?

C) Are you the truthful guard?

D) Which door should I choose?

Aptitude Questions

Q1. A number is increased by 20% and then decreased by 20%. What is the overall percentage change?

A) 0%

B) 4% increase

C) 4% decrease

D) 2% decrease

Q2. The price of a laptop is increased by 25%. By what percentage should the increased price be reduced to bring it back to the original price?

A) 15%

B) 20%

C) 22.5%

D) 25%

Q3. A student scored 360 marks and failed by 40 marks. If the passing percentage is 40%, what are the maximum marks?

A) 900

B) 1000

C) 1100

D) 1200

Q4. The population of a town increases by 10% in the first year and 20% in the second year. What is the total percentage increase?

A) 30%

B) 31%

C) 32%

D) 33%

Q5. A person's salary is increased by 15% and then by 20%. What is the equivalent single percentage increase?

A) 35%

B) 36%

C) 38%

D) 40%

Q6. In an examination, 30% of the students failed in Mathematics, 20% failed in English, and 10% failed in both. What percentage of students passed in both subjects?

A) 50%

B) 60%

C) 70%

D) 80%

Q7. A number is first decreased by 25% and then increased by 40%. What is the final percentage change?

A) 5% increase

B) 5% decrease

C) 10% increase

D) No change

Q8. The price of a product is reduced by 20%. By what percentage must the reduced price be increased to restore the original price?

A) 20%

B) 22.5%

C) 25%

D) 30%

Q9. A candidate obtained 45% marks and failed by 30 marks. Another candidate obtained 60% marks and scored 60 marks more than the passing marks. What is the maximum possible mark?

A) 500

B) 600

C) 700

D) 800

Q10. A company's profit increases from ₹80,000 to ₹1,00,000. What is the percentage increase in profit?

A) 20%

B) 22%

C) 25%

D) 30%

Q11. In a class, 60% are boys. If the number of girls is 24, what is the total number of students?

A) 40

B) 50

C) 60

D) 72

Q12. A shopkeeper increases the marked price of an article by 30% and then offers a discount of 10%. What is the effective percentage change from the original price?

A) 15% increase

B) 17% increase

C) 20% increase

D) 13% increase

Q13. If 40% of a number is 72, what is 65% of that number?

A) 108

B) 117

C) 120

D) 126

Q14. A student's marks increase from 400 to 460. What percentage of the original marks is the increase?

A) 12%

B) 15%

C) 16%

D) 20%

Q15. The population of a city decreases by 10% in the first year and increases by 10% in the second year. What is the net change?

A) No change

B) 1% increase

C) 1% decrease

D) 2% decrease

Q16. A man's income is 25% more than his expenditure. If his income increases by 20% and expenditure increases by 10%, his savings increase by what percentage?

A) 40%

B) 50%

C) 60%

D) 70%

Q17. In a school, 72% of students passed an examination. If 224 students failed, what was the total number of students?

A) 700

B) 750

C) 800

D) 850

Q18. A number is increased by 50%. By what percentage should it be decreased to get the original number?

A) 25%

B) 30%

C) 33⅓%

D) 40%

Q19. The price of petrol increases by 25%. By what percentage should a person reduce consumption so that expenditure remains unchanged?

A) 15%

B) 20%

C) 25%

D) 30%

Q20. A candidate gets 20% more marks than another candidate. If the second candidate gets 250 marks, how many marks does the first candidate get?

A) 275

B) 290

C) 300

D) 320